Smart Study Planner
A Simple Overview of the App
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I. What This App Is

Smart Study Planner is an Android app that helps students stay organized while studying. It is built for students who juggle multiple courses, assignments, exams, and study sessions all at once.

The whole idea of the app is to put everything a student needs into one place. No need to switch between a calendar, a to do list, a timer, and a notes app. It is all here in a single, friendly app.

All the data lives on the phone itself. The app does not need internet to work. Nothing is sent to a server. Everything the user types stays on their device.

II. What The App Does

In simple words, the app helps a student do these things:

- Keep a list of all their courses, with the instructor, color, schedule, and room.
- Track every assignment, exam, quiz, and project as a task with a due date and priority.
- Get a reminder on the phone 24 hours before any task is due, so the student never misses a deadline. The reminder comes as a normal phone notification, even when the app is closed.
- Run a Pomodoro study timer with focus periods and breaks, and save study sessions to see how long they actually studied.
- See charts and stats about how much they have studied, when, and what topic/course.
- Listen to background music while studying. The music keeps playing even when the timer is running. A small bar at the top of the screen lets the student pause the music anytime without stopping their session.
- Make digital flashcards for any course and study them by flipping cards.
- Track their GPA by adding the grade they expect (or got) for each course, and see what grades they need in remaining courses to hit a target GPA.
- See a live countdown to the next big exam or deadline so they always know what is coming up.
- Read short, helpful study tips when they need a little motivation.
- See a small home screen widget showing pending tasks and today's study time without opening the app.

III. Walking Through the App, Tab by Tab

The app has a bottom bar with five main tabs. Here is what each one does, in plain words.

1. Home Tab

This is the first thing the user sees after the splash screen. It is the dashboard of the whole app and packs in a lot of useful information at a glance:

- A friendly time based greeting with emoji ("Good Morning!", "Good Afternoon!", or "Good Evening!") and today's full date.
- A daily motivational quote in the greeting card (for example, "It always seems impossible until it's done." by Nelson Mandela).
- A streak chip with a fire emoji showing how many days in a row the user has studied without a gap.

- A circular progress ring showing today's task completion (for example, "0/4 done") with an encouraging line like "Let's get started!" that changes as tasks are completed.
- Three quick stat tiles: hours studied today, hours studied this week with session count, and tasks due today.
- An Exam Countdown section with live ticking countdowns (hours, minutes, seconds) to the next upcoming exams.
- A Due Today section listing every task due today, color coded by priority, with a "See all" link to a full "Due Today" screen.
- A Study Tools card grid with shortcuts to Flashcards, GPA Calculator, Study Music, and Study Tips.

2. Courses Tab

This is where the user adds their courses for the semester. The screen shows the current semester ("Spring 2026"), how many courses are enrolled, and a search bar that filters courses by name or code as the user types.

Each course gets a name, code, instructor, color, schedule, and room. Courses appear as colorful tiles showing the course code, name, professor, the meeting days/times, and the room number. There is a plus button to add a new course. Tapping a tile opens it for editing.

When adding a course, the user can pick from a palette of 20 preset colors, so each course has its own visual identity throughout the app.

3. Tasks Tab

This is the to do list of the app. The screen header shows today's date and the title "My Tasks". The user can add assignments, exams, quizzes, projects, and other tasks. Each task has a title, an optional description, a course it belongs to, a due date and time, a priority (Low, Medium, High), and a type.

Tasks are split into four sub tabs: Pending, Done (completed), All, and Overdue. The user can sort them by due date or priority, and filter them by type or by course using clean bottom sheets. Each tab remembers its own sort and filter settings.

Tapping the circle on a task brings up a colored confirmation card asking, "Mark as Complete?" or "Mark as Incomplete?". The Edit pencil opens the task for editing, and the trash icon brings up a delete confirmation.

If the user turns on the "Remind me 24h before due" toggle when adding a task, the app will send a reminder to their phone exactly 24 hours before the due date, even when the app is closed.

4. Timer Tab

This is the entry screen for the Pomodoro study timer. It shows three mode pills (Pomodoro, Short Break, Long Break), a large preview circle with time, four indicator dots, and a big "Start Studying" button that opens the full screen timer.

Below that, three stat tiles show this week's hours, today's hours, and total Pomodoros completed. A "Recent Sessions" list shows the user's last several study sessions with the date, course, type (Free Study or Pomodoro), and duration.

Tapping Start Studying opens the actual timer screen. The classic Pomodoro method is 25 minutes of focused work followed by a 5 minute break, with a longer 15 minute break after every fourth focus session. The user can adjust the work and break durations in 5 minute steps. They can also pick which course they are studying for, so the time gets saved against that course. The big circle counts down with a progress bar, and below that the screen shows the current Pomodoro count and total time studied so far.

If the user tries to leave the timer mid-session or hits Reset with progress recorded, the app shows a friendly confirmation card asking whether to save, exit without saving, or cancel.

5. Stats Tab

This is the charts and progress screen. It shows:

- Today's total study time.
- This week's total study time and the number of sessions.
- The current daily streak with a fire emoji.
- A bar chart of how many hours were studied each day of the current week (Sunday through Saturday).
- A 28-day study streak heatmap, just like the GitHub contribution graph, with three color tiers (No study, Light, Active).
- A bar list (separate bar for each course) showing how many hours went into each course over the last 30 days, color matched to each course.
- A task completion donut chart with a center percentage, broken into Completed, Pending, and Overdue.

IV. Extra Features (Reached from the Home Screen)

These are powerful tools inside the app, all reached by tapping a card in the Study Tools grid on the Home screen.

1. Study Music

A focus sound player with a clean, calming dark blue interface. The user picks an ambient sound (for example, "Sleepy Rain" or "Fireplace Holiday" under the "Rain & Water" category), adjusts the volume slider, and presses play. A live Session Time counter shows how long the music has been running.

The music keeps playing even when the user switches to a different tab, locks their phone, or starts a study timer. A small mini player bar appears across the top of every screen so the student can pause or play the music anytime without leaving what they are doing.

2. Flashcards

The user picks a course from a dropdown, then flips through digital flashcards for that course. The front shows a question, the back shows the answer. The card flips in 3D when the user taps it or hits the Flip Card button.

After flipping, the user can mark the card as Again (still learning) or Got It! (mastered). A Mastery Progress bar at the top shows how many cards in the deck have been mastered (for example, "0 / 5 mastered"). Prev and Next buttons walk through the deck, and there is also an Add Card button and a Reset Deck button.

3. GPA Calculator

The user adds each course's name, credit hours, and grade. The app calculates the running GPA and shows the breakdown course by course.

Using the "What Do I Need?" tool, the user types in their target GPA and how many credits they have left, and the app calculates what average grade they need in their remaining courses to reach that target. There is also a Clear All Courses button to start over.

4. Exam Countdown

Shows live ticking countdowns (days, hours, minutes, seconds) to every upcoming deadline. Each card lists the task name, its type, a color coded urgency label, and the remaining time. The

countdown updates every second so when an exam is one day away, the user understands the urgency.

5. Study Tips

A library of 32 hand-picked study tips, grouped into 8 topics: Focus, Memory, Exams, Time, Motivation, Notes, Rest, and Group Study. The user can flip through tips one by one with the Next Tip button or hit a random button to pull one from any topic. Every flip is animated like a 3D card flip. Each tip is short, punchy, and immediately actionable.

V. Other Features Across the App

1. Deadline Reminders (Phone Notifications)

When a task is added with the 24 hour reminder toggle turned on, the app uses Android's AlarmManager to schedule a real phone notification exactly 24 hours before the due date. The reminder still fires even if the app is fully closed and even after the phone restarts. The user does not have to keep the app open or remember to check anything.

2. Due Today Screen

Tapping "See all →" on the Home screen's Due Today section opens a dedicated screen listing every task due today, color coded by priority, with their courses, due times, and quick links back to manage tasks.

3. Sticky Mini Music Player

A mini player bar sits at the top of every screen whenever music is playing. It shows the current track and a play/pause button, so the student can keep their study music playing during a Pomodoro session and pause it anytime without going back to the music screen.

4. Home Screen Widget

A small widget the user can drag onto their phone's Android home screen. It shows the count of pending tasks and today's study time. Tapping it opens the app. It refreshes itself every 30 minutes automatically.

5. Animated Splash Screen with Orbiting Pills

When the app launches, six pill shaped feature labels (Timer, Tasks, Goals, Stats, Cards, Music) smoothly orbit around the app icon using trigonometry, while three gold dots pulse at the bottom. The pill text always stays upright as it travels around the circle, over a custom drawn background with stars, depth ellipses, a dashed orbit ring, and compass markers. After a few seconds the splash fades into the Home screen.

6. Pomodoro Study Timer with Auto-Cycling

A full Pomodoro timer that knows the difference between focus time and break time, automatically switches between them, gives a long break every fourth round, and only counts focus time as real study time when saving sessions.

7. Streak Counter

The app figures out how many days in a row the user has studied, and shows it as a fire emoji number on the Home screen and the Stats tab. This is the same idea as the streak in apps like Duolingo or Snapchat.

8. 28-Day Activity Heatmap

A grid of colored squares showing study activity for the last four weeks, just like the green squares on a GitHub profile. More hours studied means a darker square.

9. Per Course Time Tracking

The Stats tab shows a horizontal bar for each course with the total hours the user has spent studying it in the last 30 days. The bars are colored to match each course's color.

10. 3D Card Flip Animation

Both the flashcards and the study tips use a smooth 3D card flip animation when the user moves to the next card. The card spins on its Y axis.

11. Friendly Confirmation Dialogs

Instead of using Android's plain default popups, every confirm/delete dialog in the app is a custom designed colored card with an icon, a question, and color matched buttons. Different actions get different colors (blue for complete, red for delete, etc.).

VI. Conclusion

Smart Study Planner is more than just a to-do list. It is a full study companion. A student can plan their week, track every deadline, time their study sessions, listen to focus music, drill flashcards, monitor their GPA, count down to exams, and get motivated by tips and quotes, all without leaving the app and without needing internet.

The five tabs at the bottom handle the daily essentials. The Home screen shows a personalized dashboard with greeting, quote, streak, today's progress, exam countdowns, due tasks, and shortcuts to every other tool. Behind the scenes, a local database keeps everything safe on the phone, a background service keeps the music playing while the user works, and Android notifications make sure no deadline is ever missed.